

Lecture 13

Polynomials

- A polynomial is a sum of terms, where each term has a form ax^e where x is a variable, a is coefficient and e is exponent. The largest exponent of a polynomial is called its degree.

Ex -> $A(x) = 3x^{20} + 2x^5 + 4$ or $B(x) = x^4 + 10x^3 + 3x^2 + 1$

$C(x) = 5x^{1000} + 1$ (So much memory is wasted if we represent this by using array).

Polynomial Representation

```
MAX_TERMS 100
```

```
typedef struct {
```

```
    float coef;
```

```
    int expon;
```

```
} polynomial;
```

```
polynomial terms[MAX_TERMS];
```

continue...

- Consider the two polynomials $A(x) = 2x^{1000} + 1$ and $B(x) = x^4 + 10x^3 + 3x^2 + 1$.

	start A	finish A	start B		finish B	avail
coef	2	1	1	10	3	1
exp	1000	0	4	3	2	0
	0	1	2	3	4	5

Linear Data Structure

- Non-square matrix & square matrix transpose

1	2	3
4	5	6
7	8	9



1	4	7
2	5	8
3	6	9

Finding cell in 2D Array in row major matrix

Row-major address formula:

$$\text{Address} = \text{Base} + (i \times n + j)$$